Implied-Dual Calibration for Cold-Start Constrained Bidding in First-Price and VCG Auctions

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Abstract

When a new ad enters an existing auction market, an autobidding system must choose its first constrained bid before observing any feedback from that ad. We study a conditional warm-start mechanism for this first decision: reuse historical market information through admissible implied dual variables rather than through the full historical policy. In the CPC case, the constrained bid has the form $b = s(z)v + \alpha(z)C\mu$, where $s(z) > 0$, $\alpha(z) \in [0,1]$, $v = \text{CTR} \cdot \text{CVR}$, and $\mu = \text{CTR}$. Under a valid source model, the contextual surface $\Theta(z) = (s(z), \alpha(z))$ summarizes local constraint pressure and market state; otherwise it is a declared projection target. We define an observability-aware calibration objective for this surface, state the source and support conditions under which it has structural meaning, and prove first-period accounting statements linking calibration and response error to bid error, reward loss, and constraint violation. The framework covers truthful and VCG/SPA-style auctions directly and extends to first-price and hybrid auctions through a known or independently validated response layer $\Psi(z, b)$ when bid-landscape or payment-rule support is available. A BAT-shaped known-transform bridge and controlled warm-start experiments show when response-aware and guarded IDVC initializations reduce transient objective and constraint error relative to stationary reuse, and when mismatch requires abstention. Unchanged public BAT is used as a boundary audit: validation-tuned IDVC remains negative against historical stationary reuse under the public proxy replay, so we do not claim public-BAT deployment lift.

CCS Concepts

• **Information systems** → **Online advertising**; • **Computing methodologies** → *Machine learning algorithms*; • **Theory of computation** → *Algorithmic game theory and mechanism design.*

Keywords

autobidding, cold start, constrained bidding, first-price auctions, hybrid auctions, response surfaces, winning-price prediction, calibration, implied dual variables, online advertising

1 Introduction

Consider a new ad for a product category that already has auction history. Before the ad receives any impressions, clicks, or conversions of its own, the bidding system must choose a bid. It knows the new ad's value and constraint features, the auction format, and historical market data from related ads or contexts. It does not yet know whether this ad will face the same winning-price distribution as the historical campaigns. This is the cold-start decision studied in this paper: how should an autobidder turn existing market history into a first constrained bid?

The standard constrained-bidding view gives a strong baseline. Fit the optimal bidding policy on historical data, or learn a pacing controller from previous campaigns, and reuse it for the new ad. This is natural, and under stationarity it is hard to beat. The weakness is also clear: the whole historical policy can be the wrong object to transfer when the market changes. Peak hours, category shocks, or different traffic composition can move winning prices before the new ad has enough feedback to adapt.

Our proposal is to transfer a smaller object: the dual pressure behind the constrained bid. In the simplest CPC campaign, let $\mu = \text{CTR}$ and $v = \text{CTR} \cdot \text{CVR}$. In the budget-plus-CPC model of Yang et al. [25], the truthful constrained bidding rule can be written as

$$b = s\,v + \alpha\,C\mu, \qquad s > 0, \quad \alpha \in [0,1], \tag{1}$$

where C is the target CPC. The pair (s, α) is a reparameterization of the budget and CPC dual variables. Instead of replaying all historical bids, we estimate a contextual surface $(s(z), \alpha(z))$ from historical logs and use it as the first-period market state for the new ad. The surface is low dimensional, tied to the constraints, and easier to update after feedback arrives.

This shift from policy reuse to dual reuse changes what the paper must prove. First, it must define the cold-start information set, the success criterion, and the historical stationary baseline. Second, it must show how the implied dual surface is calibrated. Third, it must state when the fitted surface has structural meaning. In truthful or VCG-like auctions, the observed bid or bid surface can directly play the role of the response in Eq. (1). In first-price and hybrid auctions, the observed bid is no longer a truthful margin: it must be transformed through a response layer supported by bid-landscape or payment-rule information. With payments only, missing rule metadata, or censored advertiser-side logs, the fitted object may still be useful as a declared proxy or diagnostic, but it should not be called a recovered structural market state.

The empirical target is deliberately separated into decision and calibration evidence. The decision-level goal is early value or conversions under the same budget and CPC or ROI tolerance. The calibration-level proxy is held-out transformed calibration loss on supported data,

$$\text{TCL} = \frac{1}{n_{\text{test}}} \sum_i \left(Y_i^{\dagger} - f_i^{\top} \hat{\Theta}(z_i) \right)^2,$$

where $Y^{\dagger}$ is the truthful, transformed, or declared response for the relevant observability regime. We compare against stationary and interpretable baselines such as flat or historically reused implied-dual variables, and against flexible predictive references such as high-cardinality category-price means. The latter are expected to predict well; their role is to mark the boundary between compact calibration and unconstrained prediction.

The contributions follow the cold-start story:

Table 1: Closest lines of work and the gap addressed here.

Line	Main object	This paper
Yang/PID autobidding	Online constrained control after feedback	first-period dual state before own feedback
Unified constraints	richer bidding interfaces and KKT rules	reusable calibration target and claim boundary
Robust FPA learning	long-horizon regret or pacing under feedback	supported cold-start response inversion
Bid-landscape prediction	price or win-probability accuracy	nuisance transform for interpretable dual reuse

- We formulate cold-start constrained autobidding as a first-period decision problem with historical market data, advertiser constraints, and no own feedback from the new ad.
- We propose implied-dual calibration as the reusable market representation, starting from the concrete CPC rule in Eq. (1).
- We define which response is being calibrated under truthful, first-price, hybrid, payments-only, and censored observability regimes, so the cold-start claim is not overstated.
- We give the core support taxonomy, identification accounting, monotone inversion, and curvature diagnostics needed to distinguish structural dual calibration from response-layer and proxy effects.
- We evaluate the framework on public BAT data, a BAT-shaped known-transform bridge, and controlled warm-start stress tests, separating admissible IDVC evidence from proxy diagnostics and reporting when historical stationary reuse remains stronger.

We do not derive a new single-campaign bidding rule. The contribution is an observability-aware calibration operator for transferable contextual dual surfaces, together with support conditions that connect this object to first-period decisions. Closest work differs in role: Yang/PID-style methods derive and adapt bidding controls after feedback; unified constrained bidding expands advertiser interfaces; robust first-price work gives online guarantees under bandit feedback; bid-landscape prediction optimizes predictive accuracy. This paper asks which compact dual state can be reused before the new ad has feedback, and when that reuse has structural meaning.

2 Cold-Start Bidding and Calibration Target

This section states the reader contract before introducing the general machinery: what is known at cold start, what object is estimated from history, how success is measured, and what each observability regime allows us to claim.

Definition 1 (Cold-start constrained bidding). *A new ad enters an existing market context at period $t = 1$. Before bidding, the system observes the ad features, campaign constraints, auction format, and historical logs from related contexts, but no outcomes generated by this ad. A cold-start policy maps that information to a first bid b_1. After first-period feedback arrives, the problem becomes online adaptation or pacing rather than pure cold start.*

For decision-level evaluation, success means obtaining more early value, clicks, contacts, or conversions under the same budget and CPC or ROI tolerance. For the calibration experiments in this paper, which use public logs rather than a full online deployment, the immediate measurable target is whether the historical data can produce a compact, supported dual surface that explains held-out transformed bids or prices. This calibration target is the precondition for the first-period bid.

Let z_i denote context, such as product category, hour, device, traffic source, or a learned market embedding. Let f_i collect the value and constraint features that enter a scalar bidding margin. In the CPC case used throughout the paper,

$$f_i = (v_i, C_i\mu_i)^\top, \qquad v_i = \mathrm{CTR}_i \cdot \mathrm{CVR}_i, \qquad \mu_i = \mathrm{CTR}_i.$$

The implied-dual surface is $\Theta(z) = (s(z), \alpha(z))^\top$ in the CPC case, so the calibrated margin is $f_i^\top \Theta(z_i)$. This is the object transferred from history to the new ad. More general advertiser interfaces can add affine constraint blocks, but the main text keeps the CPC surface as the primary object.

The main stationary baseline reuses the historical optimum. In dual language, it estimates a policy or dual vector from historical data and applies the same object to the new ad. This baseline is strong when the winning-price process is stable and historical support is rich. It is weak when the first period of the new ad occurs after a shift in hour, category composition, or auction response. Implied-dual calibration keeps the useful part of that baseline—the constrained dual representation—but smooths and updates it as a contextual surface rather than replaying the full historical policy.

The response being calibrated depends on observability. We write it as

$$Y_i^\dagger = \begin{cases} b_i, & \text{truthful or VCG-like,} \\ b_i + \Psi(z_i, b_i), & \text{supported FPA/hybrid,} \\ \text{declared proxy}, & \text{unsupported or censored.} \end{cases}$$

Here $\Psi(z, b)$ is a response layer induced by first-price or hybrid payment effects and the bid landscape. It is structural only when rule metadata, threshold information, bid-landscape information, normalization, or auxiliary variation supports that interpretation.

Definition 2 (Implied-dual calibration). *Given observations $\{(f_i, z_i, Y_i^\dagger)\}_{i=1}^n$ for a declared observability regime, an implied-dual calibration fit is a regularized solution of*

$$\hat{\Theta} \in \arg\min_{\Theta \in \mathcal{A}} \frac{1}{n} \sum_{i=1}^{n} \left(Y_i^\dagger - f_i^\top \Theta(z_i) \right)^2 + \lambda_\Theta \mathcal{R}_\Theta(\Theta),$$

where $\mathcal{A}$ encodes admissibility and support restrictions, and $\mathcal{R}_\Theta$ encodes smoothness or hierarchy. The resulting first-period margin is $f_1^\top \hat{\Theta}(z_1)$ in truthful or VCG-like regimes, and $\hat{\phi}_{z_1}^{-1}(f_1^\top \hat{\Theta}(z_1))$ in first-price or hybrid regimes when a supported response transform is available. The definition specifies the estimation target; identification still depends on how $Y^\dagger$ was obtained.

The claim boundary follows the same logic. Truthful bids or a truthful bid surface support conditional dual calibration. Threshold or bid-landscape information can support the first-price transform $\Psi = G/g$ when local support and density conditions hold. Known payment rules and landscapes can support a composite hybrid response $\Psi(z, b)$, but not a unique decomposition into payment sensitivity and landscape terms without extra variation. Payments-only and censored logs support declared proxy targets or stress tests, not structural response recovery. The baseline taxonomy also

Table 2: Observability regimes and claim tags. Each experiment declares which row it occupies before interpreting IDVC.

Regime	Response used for $Y^\dagger$	Claim tag
Truthful / VCG-like	bid or truthful bid surface	structural calibration
Supported FPA / hybrid	$b + \Psi(z, b)$ from landscape or rule support	transformed structural
Payments-only BAT proxy	declared surplus proxy target	proxy / boundary replay
Censored advertiser logs	partial wins, losses, or payments	stress diagnostic
Known-transform bridge	BAT-shaped support plus injected transforms	semi-synthetic mechanism

Algorithm 1 Cold-Start Implied-Dual Calibration

Require: historical auction logs with context z_i, features f_i, bids/prices, and any rule or threshold metadata; new-ad feature vector f_1, z_1

1: Declare the observability regime: truthful, supported FPA/hybrid, payments-only proxy, or censored stress test
2: Construct the response $Y_i^\dagger$: truthful bid, supported response transform $b_i + \hat{\Psi}(z_i, b_i)$, or declared proxy
3: Fit $\Theta(z)$ by constrained smoothing:

$$\min_{\Theta \in \mathcal{A}} \sum_i \left(Y_i^\dagger - f_i^\top \Theta(z_i)\right)^2 + \lambda_\Theta \mathcal{R}_\Theta(\Theta)$$

4: Enforce dual admissibility and support restrictions; for response transforms, check monotonicity of $b + \hat{\Psi}(z, b)$ on the evaluated support
5: Produce the cold-start margin $m_1 = f_1^\top \hat{\Theta}(z_1)$ and the first bid $b_1 = m_1$ in truthful regimes or $b_1 = \hat{\phi}_{z_1}^{-1}(m_1)$ in supported FPA/hybrid regimes
6: Report $\hat{\Theta}$, the response or proxy used to create $Y^\dagger$, TCL, baseline comparisons, and cold-start diagnostics

follows this boundary: global/default feasible policies are simple deployable cold-start baselines, historical stationary reuse is the strong stationarity baseline, flat IDVC is the minimal structural baseline, contextual IDVC is the proposed compact calibrator, high-cardinality category-price means are predictive upper references rather than interpretable dual surfaces, and RL/PID/no-regret controllers are downstream adaptation baselines once feedback starts.

3 Truthful Bid Rule and Implied Duals

We start with the truthful CPC case because it explains what the new ad receives from history. Here the bid itself is the margin response, so the implied-dual surface has a direct structural interpretation and can be plugged into the first cold-start bid.

The constrained optimization problem of Yang et al. [25] is

$$\begin{aligned} \max_{0 \le x_t \le 1} \quad & \sum_{t=1}^{T} x_t \mathrm{CTR}_t \mathrm{CVR}_t \qquad (2) \\ \text{s.t.} \quad & \sum_{t=1}^{T} x_t w p_t \le B, \\ & \frac{\sum_{t=1}^{T} x_t w p_t}{\sum_{t=1}^{T} x_t \mathrm{CTR}_t} \le C. \end{aligned}$$

For nondegenerate constrained instances with $p + q > 0$, the corresponding dual variables $p, q \ge 0$ yield

$$bid_t = \frac{\mathrm{CTR}_t \mathrm{CVR}_t}{p + q} + \frac{q}{p + q} \mathrm{CCTR}_t. \qquad (3)$$

Set

$$s = \frac{1}{p + q}, \qquad \alpha = \frac{q}{p + q}.$$

Then $s > 0$, $\alpha \in [0, 1]$, and the truthful calibration surface is

$$W_{\mathrm{T}}(v, \mu, z) = s(z)v + \alpha(z)C\mu. \qquad (4)$$

The original duals map back to $p = (1 - \alpha)/s$ and $q = \alpha/s$.

The next result is the minimal identification statement behind the calibration problem: after the response has been truthfully observed or validly transformed, local rank of the features is what makes the implied-dual coefficient unique. Structural interpretation also needs a source condition on the logged policy, not just observability.

Assumption 1 (Structural response source). *In rows called structural, the observed truthful bid or transformed response is generated by a KKT-consistent constrained bidding policy for the displayed value and constraint features, up to mean-zero approximation error and cross-fitted or otherwise fixed value features. Without this source condition, the same regression identifies a behavioral or proxy projection of $Y^\dagger$, not a structural dual state.*

Proposition 1 (Conditional identification in truthful or transformed model). *Fix a context cell around z, or a shrinking neighborhood whose local approximation error is treated as part of the pseudo-true projection target, and let $f_k \in \mathbb{R}^d$ have finite second moments. Suppose the response variable $Y_k^\dagger$ is observed or validly constructed before estimating Θ_0, satisfies Assumption 1 for structural interpretation, and*

$$Y_k^\dagger = f_k^\top \Theta_0(z) + \varepsilon_k, \qquad \mathbb{E}[\varepsilon_k \mid f_k, z_k] = 0,$$

where $Y^\dagger$ is a truthful structural bid or bid surface when $\Psi = 0$, and $Y^\dagger = b + \Psi_0(z, b)$ when the first-price or hybrid response Ψ_0 is known or separately estimable. If

$$M(z) = \mathbb{E}[f_k f_k^\top \mid z]$$

is nonsingular, then

$$\Theta_0(z) = M(z)^{-1} \mathbb{E}[f_k Y_k^\dagger \mid z],$$

is uniquely identified from the transformed moment. The proposition identifies Θ_0 conditional on $Y^\dagger$; it does not identify an unrestricted Ψ_0 jointly with Θ_0. In the CPC special case, $f_k = (v_k, C\mu_k)$ and $\Theta_0(z) = (s_0(z), \alpha_0(z))$.

Proof sketch. This is conditional linear moment inversion. Multiplying by f_k and conditioning on the local context gives $\mathbb{E}[f_k Y_k^{\dagger} \mid z] = M(z)\Theta_0(z)$. Full local rank of $M(z)$ makes the affine representation in f unique after the truthful or transformed response has been constructed. The reproducibility artifact records the longer projection variants. □

If the transformed response is not exactly affine in the chosen features, the fitted surface is a projection rather than a literal dual state. The artifact records the corresponding projection, admissible-surface, spline, and local-rate variants; the main text keeps the reader focused on the calibration target and its observability conditions.

4 Response Layers for First-Price and Hybrid Auctions

We now add the response layer because first-price and hybrid auctions break the direct truthful interpretation of observed bids. The goal is to state exactly what transform would make the CPC calibration equation meaningful again.

Let D be the highest competing bid, or more generally the critical threshold, in context z, with conditional CDF G_z and density g_z. The following derivation is local to a calibrated bid interval I_z on which the threshold law is atom-free, $0 < G_z(b), g_z(b) > 0$, and the payment rule is smooth. Let $\pi(b, d, z)$ be the payment rule when the focal bidder bids b and wins against threshold $d \le b$. We assume local differentiability in b, differentiability under the integral, and boundary consistency $\pi(b, b, z) = b$ on this smooth regime. Write the normalized truthful margin as

$$m(f, z) = f^{\top}\Theta(z).$$

The impression-level dualized objective is

$$\mathcal{L}_z(b) = \int_0^b (m(f, z) - \pi(b, d, z))\, g_z(d)\, dd.$$

For an interior optimum $b^{\star} \in I_z$, Leibniz' rule and boundary consistency give

$$g_z(b^{\star})\left(m(f, z) - b^{\star}\right) = \int_0^{b^{\star}} \partial_b \pi(b^{\star}, d, z) g_z(d)\, dd.$$

Define the payment-sensitivity surface, bid-landscape surface, and composite response surface:

$$\Lambda(z, b) = \frac{\int_0^b \partial_b \pi(b, d, z) g_z(d)\, dd}{G_z(b)},$$
$$\Sigma(z, b) = \frac{G_z(b)}{g_z(b)},$$
$$\Psi(z, b) = \Lambda(z, b)\Sigma(z, b).$$

The first-order condition becomes

$$\phi_z(b^{\star}) = f^{\top}\Theta(z), \qquad \phi_z(b) = b + \Psi(z, b). \tag{5}$$

The cases are immediate. Truthful or VCG-like auctions have $\partial_b \pi \equiv 0$, so $\Psi \equiv 0$ and $b^{\star} = f^{\top}\Theta(z)$. First-price auctions have $\pi(b, d, z) = b$, so $\Lambda \equiv 1$ and $\Psi = \Sigma = G/g$. A linear hybrid rule $\pi_\lambda(b, d, z) = \lambda(z)b + (1 - \lambda(z))d$ gives $\Psi = \lambda(z)G/g$. More complex smooth hybrids can still be calibrated through the composite Ψ.

Remark 1 (Hybrid and response identifiability). *Known payment-rule metadata identify the functional form of $\partial_b \pi$, but separate identification of Λ and $\Sigma = G/g$ also requires threshold or bid-landscape information, a valid censored-landscape model, or auxiliary variation that shifts one component without the other. Without such information, the product decomposition is not unique: for any positive function $a(z, b)$, $(a\Lambda, \Sigma/a)$ yields the same composite response. Moreover, if Θ is also unrestricted, even the composite Ψ is not jointly identified from the single equation $b + \Psi(z, b) = f^{\top}\Theta(z)$ without normalizations, exclusion restrictions, or separate nuisance estimation. The empirical role of an unknown-composite $\hat{\Psi}$ is therefore diagnostic and predictive unless those additional identifying conditions are supplied.*

The next result is needed because a response transform is useful only if it can be inverted on the support being calibrated. It is an accounting statement for known or independently validated transforms; the positive FPA evidence below instantiates it with known semi-synthetic response transforms, not with an estimated production bid landscape.

Theorem 1 (FPA/hybrid fixed-point uniqueness). *Fix z and assume I_z is a supported bid interval on which $\Psi_0(z, \cdot)$ is continuously differentiable. If*

$$1 + \partial_b \Psi_0(z, b) \ge c_\phi > 0$$

for all bids in the interval, then $\phi_z(b) = b + \Psi_0(z, b)$ is strictly increasing. Hence, for every margin m in the image of $\phi_z(I_z)$, and for every interior solution when m lies in the interior image, the structural bid

$$b^{\star} = \phi_z^{-1}(m)$$

exists and is unique on I_z. If the FOC above is valid, the objective is differentiable on the interval, and the derivative has the single-crossing form $g_z(b)\{m - \phi_z(b)\}$ with $g_z(b) > 0$, this unique interior solution is the local structural optimum; margins outside the image select boundary bids under the same single-crossing condition.

Proof sketch. The derivative lower bound makes ϕ_z strictly increasing. The monotone inverse theorem gives a unique inverse on its image. The single-crossing derivative changes sign once at the inverse point, which gives the local optimum statement. □

Theorem 2 (First-period decision stability). *Fix a supported context z and feature vector f. Let $m_0 = f^{\top}\Theta_0(z)$, $\hat{m} = f^{\top}\hat{\Theta}(z)$, and suppose $\left\|\hat{\Theta}(z) - \Theta_0(z)\right\| \le \varepsilon$ and $\|f\| \le F$. In a truthful regime set $\phi_z(b) = b$; in a supported first-price or hybrid regime assume $\phi_z(b) = b + \Psi_0(z, b)$ satisfies $\phi_z'(b) \ge c_\phi > 0$ on a supported bid interval I_z and both margins lie in $\phi_z(I_z)$. Then*

$$|\hat{b} - b^{\star}| \le \frac{F\varepsilon}{c_\phi}, \qquad b^{\star} = \phi_z^{-1}(m_0), \quad \hat{b} = \phi_z^{-1}(\hat{m}).$$

If the deployed inverse uses an estimated response operator $\hat{\phi}_z$ with $\sup_{b \in I_z} |\hat{\phi}_z(b) - \phi_z(b)| \le \eta$ on the same interval, assume $\hat{\phi}_z$ is monotone on I_z, $\hat{m}$ lies in $\hat{\phi}_z(I_z)$, and $\hat{b}$ solves $\hat{\phi}_z(\hat{b}) = \hat{m}$. The bound becomes

$$|\hat{b} - b^{\star}| \le \frac{F\varepsilon + \eta}{c_\phi}.$$

Consequently, if the first-period reward curve $R_z(b)$ and cost curve $K_z(b)$ are Lipschitz with constants L_R and L_K, then

$$|R_z(\hat{b})-R_z(b^\star)| \le L_R(F\varepsilon+\eta)/c_\phi, \qquad |K_z(\hat{b})-K_z(b^\star)| \le L_K(F\varepsilon+\eta)/c_\phi.$$

If $b^\star$ satisfies a constraint threshold $K_z(b^\star) \le \kappa$, then the excess violation obeys $\max\{0, K_z(\hat{b}) - \kappa\} \le L_K(F\varepsilon + \eta)/c_\phi$.

Proof sketch. The margin error satisfies

$$|\hat{m} - m_0| \le \|f\| \left\|\hat{\Theta}(z) - \Theta_0(z)\right\| \le F\varepsilon.$$

The inverse-Lipschitz property of ϕ_z gives the bid bound. For the estimated inverse, $\hat{\phi}_z(\hat{b}) = \hat{m}$ and the uniform response error imply $|\phi_z(\hat{b}) - m_0| \le F\varepsilon + \eta$. Lipschitz reward and cost curves then give the final statements. This is a first-period stability bound, not a regret theorem. □

The theorem also gives the promised comparison with historical stationary reuse. Let b_{hist} be the bid produced by reusing the historical optimum under the same cold-start information set: historical logs, the new ad's features and constraints, and no feedback from the new ad. The historical policy may be optimal for its source market, but in the target market it pays the margin error induced by market shift.

Proposition 2 (Target-space sufficient condition against historical reuse). *Under the assumptions of Theorem 2, assume b_{hist} lies in, or is clipped to, the same supported bid interval and that the response map also satisfies $\phi_z'(b) \le C_\phi$ on that interval. Define the target first-period margin $m_0 = f^\top\Theta_0(z)$ and suppose the historical stationary policy has target-space shift at least Δ_{hist}:*

$$|\phi_z(b_{\text{hist}}) - m_0| \ge \Delta_{\text{hist}}.$$

For an IDVC initialization with $\left\|\hat{\Theta}(z) - \Theta_0(z)\right\| \le \varepsilon$, $\|f\| \le F$, and response error η, if

$$F\varepsilon + \eta < \frac{c_\phi}{C_\phi}\Delta_{\text{hist}},$$

then

$$|\hat{b} - b^\star| \le (F\varepsilon + \eta)/c_\phi < \Delta_{\text{hist}}/C_\phi \qquad \le |b_{\text{hist}} - b^\star|.$$

Thus, under the same cold-start information set, IDVC has a smaller first-period bid error than direct historical stationary reuse. The corresponding reward and cost deviations are bounded for IDVC by Theorem 2; without additional lower-sensitivity assumptions on R_z or K_z, the proposition is a bid-space comparison rather than a strict reward-ordering theorem.

Proof sketch. The IDVC bound is Theorem 2. Since ϕ_z is C_ϕ-Lipschitz from above on the supported interval, $|\phi_z(b_{\text{hist}})-\phi_z(b^\star)| \le C_\phi|b_{\text{hist}}-b^\star|$. The historical target-space shift therefore gives $|b_{\text{hist}}-b^\star| \ge \Delta_{\text{hist}}/C_\phi$. The displayed condition makes the IDVC upper bound smaller than this lower bound. □

This is the theoretical win condition against one important baseline: contextual dual calibration error plus response-transform error must be smaller than the target-market shift made by directly reusing the historical stationary policy. It is not a theorem against every simple deployable baseline; public BAT even shows that a global feasible-bin policy can be stronger than historical stationary reuse under the declared proxy replay. Controlled and known-transform settings can estimate the historical-shift condition because the target state is known; unchanged public BAT cannot, so its negative replay is reported as boundary evidence rather than as a structural falsification.

4.1 Curvature as a response diagnostic

The unified operator explains why a curved observed bid surface does not automatically imply contextual dual heterogeneity. Even with flat (s, α), the inverse response ϕ_z^{-1} can curve the surface. The diagnostic below prevents a common overinterpretation: curvature in bids is evidence of model misspecification, but not automatically evidence of contextual dual heterogeneity.

Theorem 3 (Rank-one curvature diagnostic). *Under the assumptions of Theorem 1, assume Ψ is twice differentiable in b, has no direct dependence on f except through the bid, and $f^\top\Theta(z)$ lies in the interior image of ϕ_z. For fixed z and fixed $\Theta(z)$, the response-only curvature of $b^\star = \phi_z^{-1}(f^\top\Theta(z))$ satisfies*

$$\nabla_f^2 b^\star = \chi(z, b^\star)\Theta(z)\Theta(z)^\top, \qquad \chi(z, b) = -\frac{\partial_{bb}\Psi(z, b)}{(1 + \partial_b\Psi(z, b))^3}.$$

Thus response-induced scalar-margin curvature has rank at most one. The converse is not unique: other single-index misspecifications can also produce rank-one curvature, so the result is a falsification diagnostic rather than a source-attribution theorem.

Proof sketch. Implicitly differentiate $\phi_z(b^\star) = f^\top\Theta(z)$. The first derivative is $\Theta(z)/(1 + \Psi_b)$. Differentiating once more yields the displayed Hessian. □

When the response surface is known or separately estimated from logs, the equation above suggests the generated response

$$\tilde{y}_i = b_i + \hat{\Psi}(z_i, b_i).$$

If the payment rule and bid landscape are observed or estimable, $\hat{\Psi}$ can be built from $\hat{G}, \hat{g}$, and $\hat{\Lambda}$. If the rule is unknown, $\hat{\Psi}$ may be fit directly as a composite nuisance, but then the result is diagnostic unless the missing identifying restrictions are supplied. Appendix A records the compact proof details and generated-regressor clarification used by the experiments.

5 Experiments

The empirical section answers three reviewer-facing questions. RQ1 asks what unchanged public BAT can support and where it stops. RQ2 asks whether IDVC improves first-period or early-window control when observability and support are controlled. RQ3 asks whether the response layer and mismatch guard are necessary. Tables 3 and 4 separate the public-data boundary audit from the supported mechanism evidence before the detailed results.

Public BAT audit and replay. BAT exposes public simulator surplus curves rather than raw platform winning prices, so the target is a declared proxy, not a recovered winning-price label. The calibration audit uses an IID-style period-block diagnostic split, `period // 3600 mod 5`, with fold 4 held out; this is separate from chronological campaign replay and is not presented as future-valid cold-start evidence. The audit scans 23.86M VCG rows, of which 5.72M have

Table 3: Public BAT audit: no deployment-lift claim. These rows define the public-data boundary and are deliberately separated from positive mechanism evidence.

Audit	Protocol	Main result	Claim allowed
Proxy calibration	Public BAT proxy target $Y^{\dagger}$ = WinBid/max(Count, 1); period-block IID diagnostic holdout.	VCG admissible TCL improves from 46.20 to 42.11; FPA improves only from 7182.63 to 7108.14 and all 51 FPA bins violate admissibility before projection.	proxy diagnostic
Frontier replay	Chronological 70/10/20 campaign split; scalar grid fixed before test; paired campaign bootstrap with seed 20260522.	FPA shrunk IDVC lift is -42.1% $[-44.4, -39.7]$; VCG contact lift is $+8.3\%$ $[+7.0, +9.7]$ but with larger violation. Global feasible-bin is the stronger simple BAT baseline.	boundary only

Table 4: Supported mechanism evidence: conditional decision lift. These rows use declared support or controlled response transforms and do not imply public-BAT deployment lift.

Question	Protocol	Main result	Claim allowed
Supported bridge	BAT-shaped support with frozen known response transforms; 100 seeds, 60 contexts, 6000 clusters, and 3x3 shift/response sensitivity.	Guarded IDVC beats historical stationary reuse by $+4.10\%$ $[+4.09, +4.12]$ with lower violation, passing negative controls and all sensitivity cells.	conditional lift
Mechanism ablation	Controlled warm-start suite with FPA response-bias and mismatched-prior scenarios.	Blind FPA contextual transfer is negative, response-IDVC repairs the transform, and the guard abstains in mismatched runs.	mechanism check

usable two-feature support after zero-design filtering, and 9.47M FPA rows across 51 context bins. Enforcing $s > 0$ and $0 \le \alpha \le 1$ is part of the experiment, not a cosmetic post-processing step. TCL is therefore a calibration diagnostic: it says whether the declared transformed response is well explained by a compact admissible dual surface. The stability theorem in Section 4 turns such error into decision error only when the local rank, response, and support assumptions hold.

Campaign replay answers a different question. Campaigns are sorted by `campaign_start`; the first 70% train, the next 10% validate scalar frontiers on a fixed seven-point grid from 0.50 to 3.00, and the last 20% form held-out cold-start campaigns. Per-period budgets use the declared `campaign_start`/`campaign_end` duration from the BAT campaign table, not observed test-period outcome counts. Replay reports the first 1, 3, 6, 12, and 24 observed periods, plus lifetime-share horizons, in the artifact. Campaign-bootstrap CIs include campaigns with at least one evaluated public-stats period for both the selected policy and historical comparator over the fixed horizons; this gives 2060/2060 FPA test campaigns and 20672/45330 VCG chronological test campaigns. The primary comparator is historical stationary reuse at scale 1.00; other policies may tune only on validation subject to budget violation no worse than 1.05 times the historical validation violation. The main result is a boundary result: FPA validation-tuned shrunk IDVC under-bids under the public proxy target, with fixed-horizon contact lift -42.09% and 95% campaign-bootstrap CI $[-44.35, -39.72]$. In VCG, shrunk IDVC raises contact AUC by $+8.31\%$ with CI $[+7.04, +9.69]$, but the budget-violation difference is positive, $+9.00$ with CI $[+7.76, +10.35]$, so the constrained gate fails. A simple global feasible-bin policy is stronger than historical stationary reuse on contact AUC in both auction components. These frontier results are why the paper does not claim public-BAT deployment lift.

Supported bridge and controlled decision evidence. The positive decision evidence is deliberately separated from unchanged public BAT. The BAT-shaped bridge keeps BAT-like campaign and context support but injects frozen known response transforms for matched, FPA response-bias, hybrid, abrupt-shift, null-stationary, and unsupported-mismatch regimes. It is therefore semi-synthetic mechanism evidence, not public-BAT deployment lift. The primary bridge contrast compares guarded IDVC with historical stationary reuse across the four supported shift regimes. Over 100 seeds and 60 contexts, guarded IDVC improves fixed-horizon early-AUC by 4.10% with 95% cluster-bootstrap CI $[4.09, 4.12]$, while reducing violation area by 0.017; null-stationary and unsupported-mismatch negative controls pass, and the guard abstains in all unsupported-mismatch contexts. The 3x3 sensitivity grid over shift and response multipliers also passes, with the smallest lower confidence bound still positive at $+3.37\%$.

The complementary controlled warm-start suite uses matched stationary markets, smooth drift, abrupt shifts, mismatched priors, FPA response bias, and hybrid payment interpolation. Every policy uses the same stable residual controller after initialization, so differences isolate the first state supplied to the controller. The strongest deployable comparator is chosen within each setting from historical stationary reuse and global/flat IDVC. Uncertainty is computed over 100 seeds.

Mechanism ablation. Figure 2 is the compact ablation view. The first panel shows why FPA needs the response layer: blind contextual IDVC transfers the truthful margin into the wrong bid space and loses early reward, while response-IDVC inverts the supported response and recovers a positive lift. The second panel shows why mismatch guarding is not optional: the guard abstains in all mismatched-prior runs and replaces unsafe contextual transfer

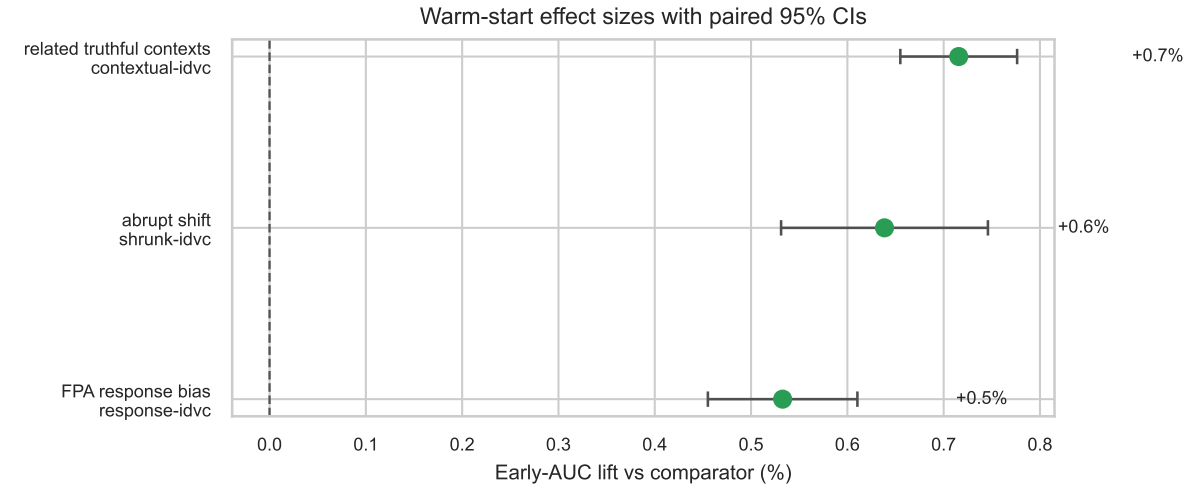


Figure 1: Warm-start effect sizes. Each point is the Early-AUC lift of the stated IDVC variant against the strongest deployable non-oracle comparator, with a seed-level 95% confidence interval and a zero-effect reference line. The takeaway is modest but consistent positive mechanism evidence under controlled support, not a claim of public-BAT deployment lift.

Table 5: Decision-level warm-start effects over 100 seeds. Early-AUC is average cumulative reward over 100 controller updates; negative time-to-target and violation changes are improvements. The comparator is the strongest deployable historical/global baseline available for that scenario. Mismatch is reported as a guard ablation in Figure 2, not as a main lift row.

Setting	IDVC variant	Comparator	Early-AUC lift (95% CI)	TTT change	Violation change
Related truthful contexts	Contextual IDVC	best of historical/global	+0.72% [+0.65, +0.78]	−3.0	−0.060
Abrupt shift	Guarded IDVC	best of historical/global	+0.64% [+0.53, +0.75]	−2.1	−0.068
FPA response bias	Response-IDVC	best of historical/global	+0.53% [+0.46, +0.61]	−1.2	−0.020

with the conservative global prior. The third panel keeps the public BAT boundary visible beside the mechanism result: calibration loss can improve while unchanged replay remains negative, and decision lift is shown only when the predeclared BAT gate passes.

The remaining synthetic checks in the artifact test monotone inversion, rank-one curvature, hybrid non-identifiability, generated regressors, censoring, and weak feature rank. They support the same claim boundary: richer response-aware calibration is meaningful only under explicit support and observability assumptions. Public advertiser-side RTB logs remain censored-observability stress tests rather than evidence for platform-side response identification.

6 Related Work

This section is placed after the calibration problem and main evidence so the literature is read as positioning rather than as a prerequisite for understanding the paper.

Autobidding with constraints. Constrained autobidding has been studied from both single-agent and market-level perspectives [2, 4, 17, 25]. Yang et al. [25] derive the dual formula used as our truthful baseline and propose model-predictive PID updates. Multi-constraint bidding systems generalize this idea by deriving unified optimal bidding functions for richer advertiser constraints [12]; recent pacing work compares separate, min-pacing, and fully coupled dual updates for budget and return-on-spend constraints [3]. Lucier et al. [17] give robust bidder-side guarantees under bandit feedback and support first-price, second-price, and intermediate formats. These works primarily address online policy design once bidding feedback is available. Our work asks a different question: before a new ad has its own feedback, what compact dual state can be reused from historical market logs?

Benchmarks and first-price bidding. BAT [14] provides an autobidding benchmark with both VCG and first-price auction components, which makes it a natural empirical anchor for the split between truthful calibration and response-aware diagnostics. Recent work on first-price pacing, non-truthful autobidding, first-price learning with budgets, and bid shading studies robust or prior-independent strategies when the bidder must reason about the competing-bid distribution [6, 15, 16, 18, 19, 21]. We use this literature differently: the platform-side distribution G_z and payment sensitivity are nuisance objects that help transform observed bids into scalar margins when the observability regime supports doing so. The comparison point is not that calibration replaces robust online bidding, but that it can initialize or regularize such bidding in the cold-start period.

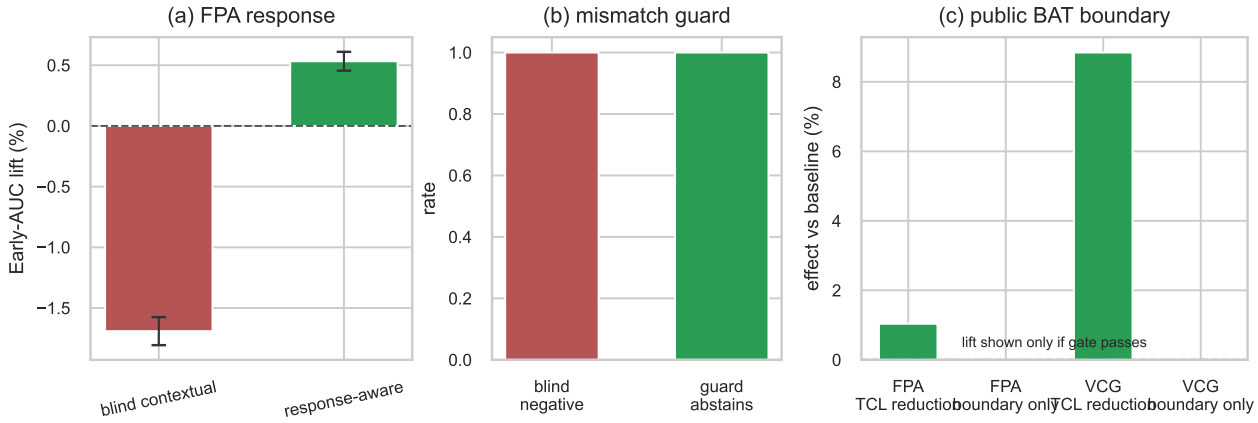


Figure 2: Mechanism and limits. Panel A asks whether the response layer matters in FPA; response-aware inversion turns negative blind transfer into positive early lift. Panel B asks whether transfer should ever abstain; the mismatch guard abstains exactly in the designed mismatch regime. Panel C contrasts calibration diagnostics with unchanged BAT replay, showing why BAT is boundary evidence rather than a deployment-lift result.

Winning-price and bid-landscape prediction. Winning-price prediction in real-time bidding is often modeled through censored regression, survival analysis, or neural bid-landscape forecasting [20, 22–24, 26]. These methods optimize predictive accuracy directly. Our use of bid landscapes is narrower: when the observability regime supports the transform, the landscape is a nuisance input for interpretable dual calibration; when it does not, the fitted response is only a diagnostic or predictive proxy.

Finance-style calibration. Implied volatility is a latent parameter obtained by inverting an option-pricing model [5, 13]. The finance literature then moves from constant volatility to smiles, local volatility, arbitrage-aware surface fitting, neural smoothing, and operator smoothing [1, 7–11]. We borrow the calibration workflow, not the economic assumptions. Auction logs do not inherit replication, no-arbitrage, or a unique latent state from finance; the safe analogues are declared operators, admissible smoothing, response diagnostics, and support-aware claim boundaries.

7 Extensions and Limitations

This section collects what the paper deliberately does not make central: broader constraint interfaces, deployment use, and unsupported observability regimes.

Multi-constraint interfaces. The same calibration view applies when the advertiser interface exposes several affine constraint blocks that enter a scalar margin with compatible signs and a common normalization,

$$f^\top \Theta(z) = s(z)\psi_0 + \sum_{j=1}^{m} \alpha_j(z)\psi_j,$$

$$s(z) > 0, \qquad \alpha_j(z) \geq 0, \qquad \sum_{j=1}^{m} \alpha_j(z) \leq 1.$$

The exact feature blocks ψ_j are product-interface-specific, and componentwise interpretation requires local rank and stability diagnostics. We therefore treat the CPC case as the main empirical setting and the multi-constraint form as an extension.

Deployment. The calibrated surfaces are not meant to replace advertiser-side pacing algorithms. They can initialize or regularize downstream controllers, for example by bidding through $b_t = \hat{\phi}_{z_t}^{-1}(g_h f_t^\top \hat{\Theta}_h(z_t) + \delta_h)$ with a slow residual controller. The supported bridge and controlled warm-start checks in Section 5 support this use as an initialization mechanism, not as a new universal online-learning algorithm. The public BAT frontier replay does not show end-to-end lift for admissible IDVC; it shows conservative under-spending under the declared proxy mapping and, for VCG, no primary gate pass against historical stationary reuse. A production experiment or a richer BAT policy replay with a validated response transform would still be needed to claim deployment lift.

Claim boundaries. The central claim is deliberately narrow: once the observability regime supplies a truthful response or valid transform, platform logs can calibrate and smooth an implied-dual surface $\Theta(z)$. In non-truthful auctions, a response surface $\Psi(z, b)$ is structural only under extra bid-landscape, rule-metadata, normalization, or auxiliary-variation assumptions. Logs without those ingredients support declared transforms or regularized composite response targets for prediction and diagnostics, but not unique decomposition of $\Psi = \Lambda\Sigma$. Censored advertiser-side logs stress-test degradation and support overlap; they do not identify the full response.

8 Conclusion

This paper studies cold-start constrained autobidding: the first bid for a new ad on an existing market, before the ad has its own feedback. Starting from the concrete CPC rule $b = s(z)v + \alpha(z)C\mu$, we estimate compact contextual implied-dual surfaces and use them

as reusable market state. The truthful case gives the cleanest interpretation; first-price and hybrid cases require a supported response layer; payments-only and censored regimes require declared proxy targets and weaker claims.

On public BAT, admissible contextual IDVC improves over admissible flat IDVC in BAT VCG-component proxy diagnostics, while the validation-tuned frontier replay is negative against historical stationary reuse. That failure is part of the result: unchanged BAT defines the public-data boundary and prevents a deployment-lift claim. The BAT-shaped known-transform bridge and controlled warm-start experiments fill the mechanism gap: with support declared, response-IDVC repairs blind FPA transfer and guarded IDVC improves early control while abstaining under mismatch. The intended conclusion is therefore bounded: implied-dual surfaces are useful because they are compact, interpretable, and theoretically connected to first-period decisions under explicit support conditions, not because public BAT alone proves a universal FPA cold-start policy.

A Proofs and Protocol Details

Conditional identification. For Proposition 1, condition on a context cell or a shrinking local neighborhood around z and treat any local approximation error as part of the projection target. The maintained transformed moment is $Y^\dagger = f^\top \Theta_0(z) + \varepsilon$ with $\mathbb{E}[\varepsilon \mid f, z] = 0$. Multiplying by f and conditioning gives $\mathbb{E}[fY^\dagger \mid z] = \mathbb{E}[ff^\top \mid z]\Theta_0(z)$. If $M(z) = \mathbb{E}[ff^\top \mid z]$ is nonsingular, the coefficient is unique and equals $M(z)^{-1}\mathbb{E}[fY^\dagger \mid z]$. If the response is payments-only, censored, or built from an unsupported proxy, the same algebra identifies only the declared projection, not a structural dual state.

FPA and hybrid inversion. For Theorem 1, local differentiability and boundary consistency give the first-order condition

$$g_z(b)\{m - b\} = \int_0^b \partial_b \pi(b, d, z) g_z(d)\, dd,$$

hence $m = \phi_z(b) = b + \Psi(z, b)$. If $1 + \partial_b \Psi(z, b) \geq c_\phi > 0$ on the supported interval, ϕ_z is strictly increasing and maps the interval one-to-one onto its image. Therefore every supported margin has a unique inverse bid. When the derivative of the objective has the single-crossing form $g_z(b)\{m - \phi_z(b)\}$ with $g_z(b) > 0$, that inverse is the unique local optimum; margins outside the image choose the corresponding boundary. Known FPA gives $\Psi = G/g$, while a linear hybrid payment gives $\Psi = \lambda G/g$.

First-period stability. For Theorem 2, Cauchy–Schwarz gives $|\hat{m} - m_0| \leq \|f\| \left\|\hat{\Theta}(z) - \Theta_0(z)\right\| \leq F\varepsilon$. The inverse of a map with derivative at least c_ϕ is $1/c_\phi$-Lipschitz on the supported image, so $|\hat{b} - b^\star| \leq F\varepsilon/c_\phi$. If the deployed inverse uses $\hat{\phi}$ with $\sup_b |\hat{\phi}(b) - \phi(b)| \leq \eta$, insert and subtract $\hat{\phi}(\hat{b})$ before applying the same inverse-Lipschitz argument, giving $(F\varepsilon + \eta)/c_\phi$. Lipschitz reward and cost curves then translate bid error into first-period reward and constraint-error bounds. This is not a regret guarantee after feedback starts.

Curvature diagnostic. For Theorem 3, differentiate the implicit equation $\phi_z(b^\star) = f^\top \Theta(z)$. The first derivative is $\nabla_f b^\star = \Theta(z)/(1 + \Psi_b)$. Differentiating again with respect to f gives

$$\nabla_f^2 b^\star = -\frac{\Psi_{bb}}{(1 + \Psi_b)^3} \Theta(z)\Theta(z)^\top.$$

Thus response-only curvature is rank at most one for a scalar-margin model. The diagnostic is falsifying rather than identifying: rank-one curvature can also arise from other single-index misspecifications, and higher-rank curvature does not by itself say which nuisance is wrong.

Generated response targets. When Ψ is estimated before fitting Θ, the calibration response is $\tilde{y}_i = b_i + \hat{\Psi}(z_i, b_i)$. Cross-fitting separates nuisance estimation from the dual-surface fit. If

$$\sup_i \left|\hat{\Psi}^{(-k(i))}(z_i, b_i) - \Psi_0(z_i, b_i)\right| = o_p(1)$$

on supported folds and the local Gram matrices remain nonsingular, the fitted $\hat{\Theta}$ converges to the same transformed projection target as if Ψ_0 were known, up to the nuisance error term already present in Theorem 2. Same-fold nuisance fitting can reduce apparent TCL without improving structural calibration, so the experiments treat such fits as diagnostics unless support and cross-fitting are explicit.

BAT protocols. BAT calibration and BAT campaign replay use different splits because they answer different questions. Calibration uses the period-block key

$$\lfloor \texttt{period}/3600 \rfloor \bmod 5,$$

with fold 4 held out, and reports TCL for

$$Y^\dagger = \text{WinBid}/\max(\text{Count}, 1).$$

This period-block split is an IID-style proxy diagnostic rather than chronological cold-start evidence, and no claim treats it as future-valid replay. Campaign replay uses chronological campaign-start order: 70% train, 10% validation, and 20% held-out campaigns. Per-period budget is the auction budget divided by declared planned duration from campaign start and end timestamps; the replay does not divide by the observed number of test periods or other outcome counts. Fixed horizons are first 1, 3, 6, 12, and 24 observed periods; lifetime horizons are first 5%, 10%, and 20% of observed campaign life. Deployable policies choose a scalar from the predeclared seven-point grid from 0.50 to 3.00 on validation, subject to budget-violation AUC no worse than 1.05 times historical stationary reuse at scale 1.00. Test uncertainty uses paired campaign-level bootstrap confidence intervals with seed 20260522 and 1000 resamples. The public replay is unchanged logged-data boundary evidence, not a tuned online deployment simulator.

Warm-start protocol. The BAT-shaped bridge uses 100 seeds, 60 contexts, fixed known transforms, and cluster bootstrap over synthetic campaign/context clusters. Its uncertainty is conditional on the frozen scenario registry and is not empirical BAT uncertainty. The registry enumerates matched, FPA response-bias, hybrid, abrupt-shift, null-stationary, and unsupported-mismatch regimes before the full run; the artifact records the numerical shift and response parameters. The bridge must also pass a 3x3 sensitivity grid over shift and response multipliers. The controlled suite runs 100 seeds and 100 controller updates per seed. The controller after initialization is held fixed, so the measured effect is the initial

state supplied by historical stationary reuse, global/flat IDVC, contextual IDVC, response-IDVC, or guarded IDVC. Early-AUC is the seed-level average cumulative reward over those controller updates. Time-to-target is the first update where the constraint gap remains within tolerance, and violation change is the seed-level change in early violation area. Confidence intervals in Table 5 and Figure 1 are computed over seeds.

Artifact and Reproducibility Statement

An anonymized artifact bundle submitted with the paper contains the frozen protocol, BAT split logic, preprocessing scripts, column mapping, scalar-grid frontier selection, bootstrap seeds, row counts, generated tables and figures, and the exact scripts used to recreate the reported public-data diagnostics. BAT calibration uses the fixed period-block key $\lfloor \texttt{period}/3600 \rfloor$ mod 5 with fold 4 held out as an IID proxy diagnostic. BAT campaign replay uses the chronological `campaign_start` split described in Appendix A; the full frontier run is uncapped, and capped runs are smoke tests only. The supported bridge uses seed 20260522, 100 seeds, 60 contexts, the declared known-transform scenario registry, and the 3x3 shift/response sensitivity grid. The controlled warm-start suite reports 100 seeds.

GenAI Usage Disclosure

ChatGPT and Codex were used to assist with manuscript critique, revision planning, writing polish, experiment-code assistance, and figure/table generation. The authors reviewed, edited, and manually verified citations, mathematical statements, proofs, code, data-processing outputs, numerical results, and final text; no citations, data, or results were accepted solely from GenAI output. No private or non-public review materials were used in the preparation of this disclosure.

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